

A DIGITAL THESIS

Neural Debris Removal — *forensic recovery of a poisoned sky*

*On locating and neutralizing a data-poisoning backdoor in an
astronomical streak detector, without deleting a single
legitimate detection.*



Fig. 0 — Live detection field, channel OBS-7.02. Boxes lock onto streaks as the scan crosses them; poison is flagged in real time.

AUTHOR

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DIVISION

Observatory · Adversarial ML

SUBJECT MODEL

STATUS

● **NDR** · a digital thesis

Relational DEF. EPN I. Breach II. Related Background III. Method IV. Apparatus V. Experiment

DEFENSE

SEED

Unlearning ensemble, ×4

42

CHAPTERS

READING TIME

IX + references

~14 min

BEGIN READING ↓

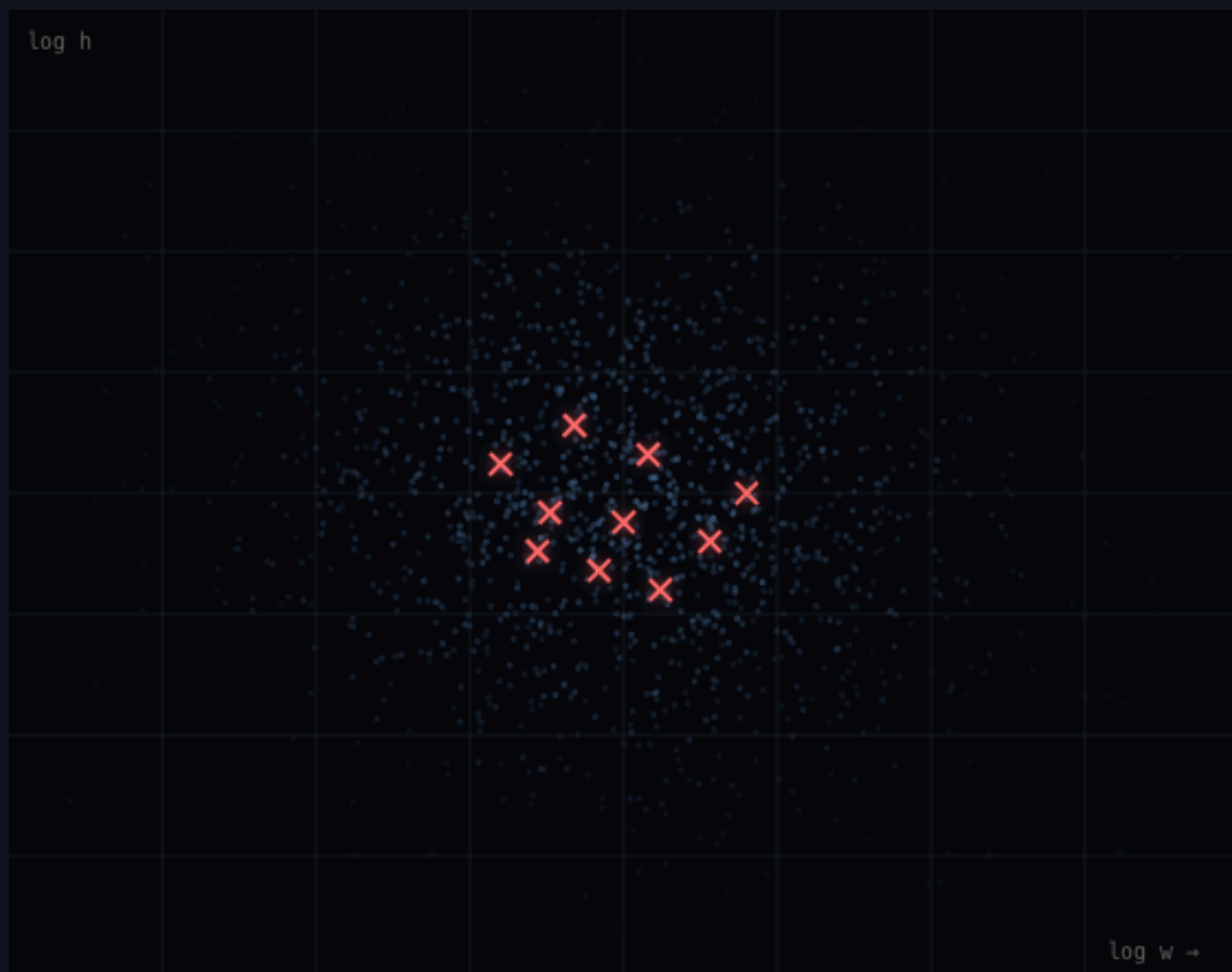


Fig0 candidate distribution, log-geometry space. Blue points are every box the poisoned model proposes on the test field; red crosses mark known noise around truth. The poison sits mid-distribution, indistinguishable from legitimate candidates by shape alone.

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II. Related

III. Method

IV. Apparatus

V. Experiment

note

Spent the better part of a week engineering geometry features before running this plot. The distribution killed all of them in about ten seconds. The lesson stuck — look before you engineer.



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II. Related

III. Method

IV. Apparatus

V. Experiment



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II. Related

III. Method

IV. Apparatus

V. Experiment



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I. Breach

II. Related

III. Method

IV. Apparatus

V. Experiment

3.3

Unlearning ensemble – classifier-only fine-tune

The model is fine-tuned on the unlearn set with *empty* labels, but with everything frozen except the classification head (`cls_score` , `cls_subnet`). The detector keeps its eyes; it forgets only what it was told to shout about. Four seeds and learning rates form a disagreement committee.

model	lr	iters · seed
ens · A	3e-4	20 · s42
ens · B	5e-4	20 · s42
ens · C	5e-4	30 · s7
ens · D	7e-4	15 · s13

frozen: **backbone** · **FPM** · **box regression** · **anchors** – only classification logits move.

note

Why cell 10 exists: a portable trigger means I get to calibrate on my own synthetic set. A contextual trigger means I keep my hands off the local knobs entirely.

3.6

Confidence remap – the guidance program

The three signals fuse into `p_poison` (Eq. 2). High `p_poison` on a strong box is *demoted to* ε rather than deleted; low `p_poison` is trusted as-is; the region between ramps down; noise-floor boxes are dropped outright (Eq. 3). This surface is built as an interactive apparatus in Chapter IV.

$$p_{\text{poison}} = 0.75 \text{ signal}_{\text{surv}} + 0.20 \text{ signal}_{\text{app}} + 0.05 \text{ signal}_{\text{geo}}$$

(2)

$$\text{remap}(s, p) = \begin{cases} 0 & s < \text{MIN_KEEP} \\ \varepsilon & p \geq P_{HI} \\ s \text{ (or } C_{\text{max}} \text{ if BOOST)} & p \leq P_{LO} \\ \max\left(\varepsilon, s\left(1 - \frac{p - P_{LO}}{P_{HI} - P_{LO}}\right)\right) & \text{otherwise} \end{cases}$$

(3)

P_HI flag $\rightarrow \varepsilon$

P_LO trust

MIN_KEEP drop noise

ε 0.01



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II. Related

III. Method

IV. Apparatus

V. Experiment

note

Deleting flagged boxes felt satisfying and scored terribly. Demoting to ε preserves the box geometry the metric secretly rewards – the hardest lesson of this thesis.



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I. Breach

II. Related

III. Method

IV. Apparatus

V. Experiment



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I. Breach

II. Related

III. Method

IV. Apparatus

V. Experiment



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I. Breach

II. Related

III. Method

IV. Apparatus

V. Experiment



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I. Breach

II. Related

III. Method

IV. Apparatus

V. Experiment



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I. Breach

II. Related

III. Method

IV. Apparatus

V. Experiment



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I. Breach

II. Related

III. Method

IV. Apparatus

V. Experiment



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I. Breach

II. Related

III. Method

IV. Apparatus

V. Experiment



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I. Breach

II. Related

III. Method

IV. Apparatus

V. Experiment



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II. Related

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